

A reproducible benchmark for transformation uncertainty in aquifer-test parameter inference

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ABSTRACT

Aquifer-test transmissivity and storage are not direct observations. They are inferred by fitting a response model to drawdown. This paper introduces a reproducible computational benchmark that quantifies how that drawdown-to-parameter transformation affects inferred aquifer properties before they enter groundwater models, pumping-capacity calculations, or response-time estimates. Four analytical interpretations—Theis confined flow, Hantush–Jacob leakage, lagging Darcy without leakage, and lagging Darcy with leakage—are fitted to controlled numerical responses and two field pumping tests. A 10,000 scenario groundwater-flow ensemble generated 160,012 fitted interpretation records across homogeneous, leaky, finite-boundary, heterogeneous, and combined settings. The benchmark links support-matched reference parameters, analytical limiting checks, grid sensitivity checks, and scenario-conditioned model factors. Analytical verification confirmed physical consistency in 22 of 25 limiting cases, and numerical responses aligned with analytical references in 30 of 32 configurations. Benchmark-transfer regressions predicted the 90th percentile absolute log model factor with cross-validated R^2 values of 0.48, 0.51, and 0.55 for transmissivity, storage, and response time. Application to the Massachusetts fractured-bedrock and Lovelock Valley basin-fill cases produced benchmark-conditioned model-factor COV values of 0.61–0.90 and 0.31–0.51, respectively. Model-averaged decision diagnostics showed that the selected interpretation can understate hydraulic capacity and response-time spread. The workflow evaluates lagging Darcy as one candidate interpretation and reports transformation uncertainty through response residuals, benchmark support, field applicability, and inherited decision consequences.

1. Introduction

Groundwater models, capture calculations, dewatering designs, geothermal screening, and water-management assessments commonly inherit transmissivity and storage from pumping-test reports (Freeze and Cherry, 1979; Bear, 1979; Anderson et al., 2015). Those quantities are not measured directly. They are inferred by transforming observed drawdown through an interpretation model, such as Theis confined flow, Hantush–Jacob leakage, delayed-yield, wellbore-storage, or generalized radial-flow solutions (Theis, 1935; Cooper and Jacob, 1946; Hantush and Jacob, 1955; Boulton, 1954; Papadopoulos and Cooper, 1967; Neuman, 1975; Barker, 1988). This front-end transformation can introduce model-form error before stochastic groundwater modeling, Bayesian model averaging, hydraulic tomography, or management optimization begins (Rojas et al., 2008; Tsai, 2010; Tsai and Elshall, 2013; Doherty, 2015).

Mature groundwater uncertainty methods usually operate after a model ensemble, parameterization, or prior structure has already been specified. Slug-test support studies, pumping-test upscaling, aquifer-scale inference, and multimodel groundwater prediction address related downstream questions (Beckie and Harvey, 2002; Manewell et al., 2023, 2024; Naderi and Gupta, 2023; Li and Tsai, 2009). Pumping-test interpretation begins one step earlier, when

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a drawdown record is converted into apparent transmissivity, storage, leakage, or response-time parameters. If the response model omits temporal memory, leakage, boundary behavior, or unresolved support effects, the mismatch can be hidden inside fitted aquifer parameters and then mistaken for hydrogeologic heterogeneity.

This paper develops a reproducible benchmark for that upstream transformation step. Four analytical interpretation pathways form controlled drawdown-to-parameter mappings: Theis confined, Hantush–Jacob leaky, lagging Darcy without leakage, and lagging Darcy with leakage. Lagging Darcy enters the analysis as one diagnostic interpretation for temporal response structure within the conventional aquifer-test solution set. The computational contribution is a 10,000 scenario MODFLOW 6 benchmark containing controlled transient responses, support-matched reference parameters, analytical limiting checks, grid-sensitivity checks, and scenario-conditioned model factors. The benchmark is then linked to two public field data sets, a Massachusetts fractured-bedrock test and the Lovelock Valley MWP2 basin-fill test, through response diagnostics and model-averaged decision quantities (Cho et al., 2004, 2008; Nadler, 2020a,b).

2. Methods

2.1. Transformation uncertainty framework

The analysis treats aquifer-test interpretation as a transformation model. The measured input is a drawdown response $s_{\text{obs}}(t, r)$, and the derived outputs are apparent aquifer parameters θ_M , including transmissivity T , storage S , leakage factor B , or lagging response times under interpretation model M . For a fitted response $s_M(t, r; \theta_M)$, the response-level log model factor is

$$\eta_{M,i,j} = \log[s_{\text{obs},i}(t_j) + c] - \log[s_{M,i}(t_j; \theta_M) + c], \quad (1)$$

where i denotes an observation well, j denotes time, and c is a small positive offset used only to keep the log transformation defined. The dispersion SD_e of η quantifies response-level transformation residual structure. For numerical benchmark cases with known support-matched reference values, parameter model factors are also computed, for example $M_T = T_{\text{app}}/T_{\text{ref}}$ and $M_S = S_{\text{app}}/S_{\text{ref}}$. Lognormal coefficient of variation values are computed from the standard deviation of $\log M$.

2.2. Field data and analytical interpretations

The field component uses two public pumping-test data sets with contrasting hydrogeologic settings and observation geometries. The Massachusetts case is a fractured crystalline bedrock pumping test with 12 retained observation wells and 420 drawdown observations (Cho et al., 2004, 2008). The retained diagnostic subset is analyzed as a constant-

rate response with $Q = 1.2 \times 10^{-4} \text{ m}^3 \text{ s}^{-1}$, while the original variable-rate history remains an applicability condition. The Lovelock Valley case is USGS multiwell pumping test 2 in basin-fill sediments of the lower Humboldt River Basin, with pumping and recovery represented through superposition (Nadler, 2020a,b).

Both field cases are fitted using four analytical interpretations. Their confined flow is the classical no-leakage reference. Hantush–Jacob is the conventional leaky-aquifer reference. Lagging Darcy without leakage introduces flux and storage-release lag times to diagnose temporal memory, and lagging Darcy with leakage combines that temporal response with leakage. The lagging solutions are evaluated in the Laplace domain and inverted using the Stehfest algorithm (Stehfest, 1970). Fits minimize log-response residuals using the same well coordinates, observation records, and pumping representation for all interpretations. AIC, BIC, local parameter correlation, boundary-hit checks, and time-window residual summaries are retained as complexity and identifiability diagnostics.

2.3. Numerical groundwater-flow benchmark

The controlled benchmark uses MODFLOW 6 as a reproducible groundwater-flow response generator (Langevin et al., 2017; Bakker et al., 2016). In this role, MODFLOW 6 supplies transient drawdown responses for testing how analytical interpretations recover apparent aquifer parameters. The ensemble contains 10,000 scenarios in ten classes: homogeneous confined flow, distributed leakage, finite no-flow boundary, constant-head boundary, heterogeneous hydraulic conductivity, anisotropic heterogeneous conductivity, heterogeneous storage, heterogeneous leakage, heterogeneous boundary, and combined leakage and boundary settings. Each class contributes 1000 scenarios.

The design spans grid spacings of 5.0–10.0 m, square domains of 405–1205 m, six observation layouts with 3–5 wells, logarithmically spaced records over 1–10 days, and water-level noise from 0 to 0.01 m. Heterogeneous cases use correlated lognormal hydraulic conductivity or storage fields. A scenario-specific effective pumping support radius defines T_{ref} and S_{ref} before fitting. The support radius combines hydraulic diffusivity, pumping duration, maximum observation distance, leakage length where present, and boundary or domain limits. This prevents a numerical cell value from being treated as the same object as a pumping-test parameter.

Each numerical response is refitted with the same four analytical interpretations. The resulting records are screened by removing active boundary-flag fits and retaining interpretations within $\Delta BIC \leq 10$ of the best interpretation for the same response. Analytical limit checks test whether the implemented formulas recover known limiting behavior. Numerical consistency checks compare MODFLOW-generated homogeneous and leaky responses against analytical reference responses across grid spacings.

2.4. Benchmark transfer and decision diagnostic

The numerical benchmark contains controlled reference values, and the field cases contain real response diagnostics. Benchmark-transfer regressions estimate the magnitude of parameter model factors that a field test may inherit

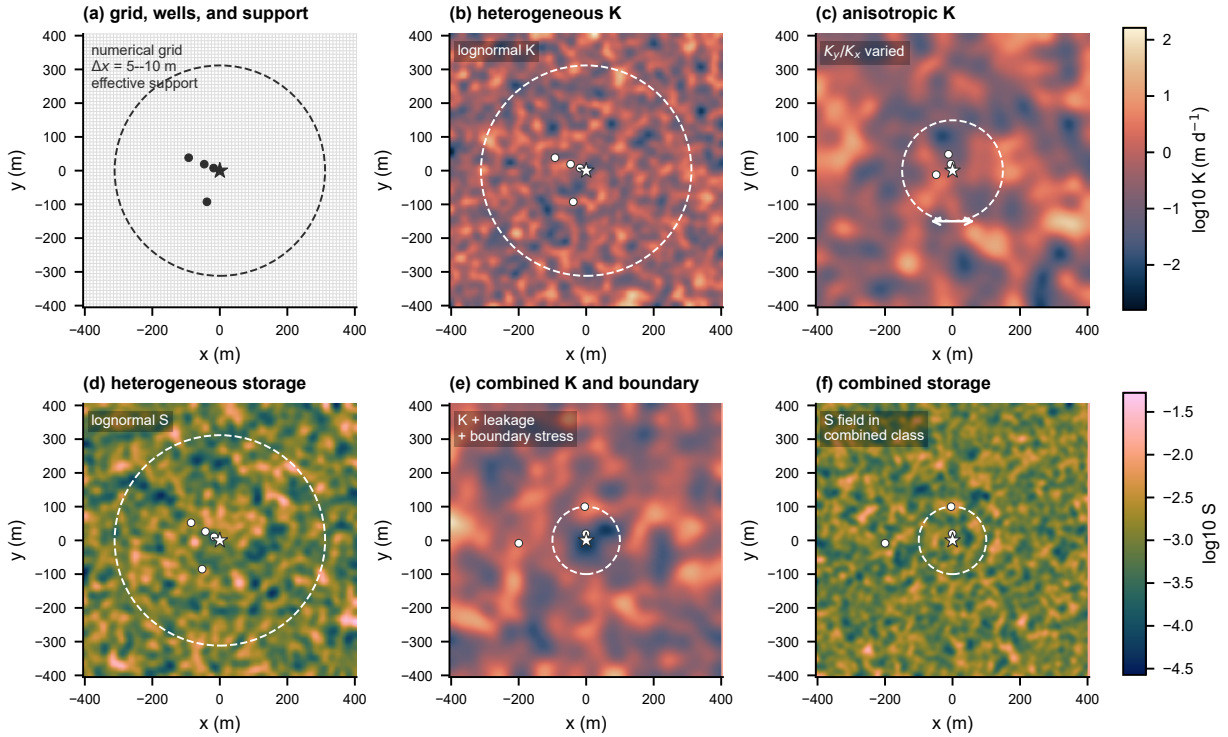


Figure 1: Numerical groundwater-flow benchmark design with MODFLOW 6 through (a) the finite-difference grid, extraction well, observation wells, and scenario-specific effective support envelope for the aquifer-test reference; (b) a representative heterogeneous hydraulic conductivity field; (c) a representative anisotropic hydraulic conductivity field; (d) a representative heterogeneous storage field; (e) combined leakage, boundary, hydraulic conductivity, and storage conditions for transformation stress; and (f) paired numerical response classes for the four analytical interpretations. The numerical fields define controlled drawdown responses for interpretation by the four analytical solutions.

when its diagnostics resemble parts of the numerical ensemble. Predictors are restricted to quantities available in both numerical and field tests: response residual dispersion, observation distance, BIC support, boundary and leakage flags, parameter spread, and fitted pathway class. Regression targets are the 90th percentile absolute log model factors for T , S , and response time within each numerical scenario.

The management diagnostic follows the Bayesian model averaging logic used in groundwater uncertainty analysis (Hoeting et al., 1999; Li and Tsai, 2009; Tsai and Elshall, 2013). The four aquifer-test interpretations are treated as candidate models. Their fitted T and S values are propagated into hydraulic capacity, proportional to T , and characteristic response time, proportional to S/T . Strict BIC weights and a variance-window weighting scheme are reported. This diagnostic asks whether a later groundwater calculation can inherit overconfident inputs if aquifer-test interpretation uncertainty is collapsed into one fitted parameter set.

Table 1

Numerical groundwater-flow benchmark for field-inference scope via MODFLOW 6 as the controlled response generator, scenario-conditioned transformation model factors for the four analytical interpretations, and field conditions under which benchmark factors can inform field diagnostics.

Component	Numerical benchmark result	Field interpretation
Coverage	10,000 numerical scenarios in ten classes; six observation layouts; 160,012 fitted interpretation and well records; 88,106 screened records.	Controls response generation before field evidence is interpreted.
Verification	Analytical limit verification confirmed physical consistency in 22 of 25 cases; numerical consistency checks aligned with analytical reference responses in 30 of 32 cases; remaining warnings identify near well leakage and coarse grid limits.	Numerical support and discretization effects remain visible and should be reported.
Model factors	Screened class and interpretation median COV values are 2.06 for transmissivity, 1.27 for storage, and 1.21 for response time; upper tails remain scenario dependent.	Conditional model factor COV values apply to matching benchmark classes.
Applicability	Massachusetts receives a primary conservative match for combined leakage and boundary behavior; Lovelock receives primary matches for heterogeneous storage and hydraulic conductivity.	Field geometry and true support scale define the applicable range of the scenario-conditioned values.

3. Results and Discussion

3.1. Benchmark verification and model factors

The numerical benchmark generated 160,012 fitted interpretation and well records, of which 88,106 satisfied the screening criteria (Table 1). Analytical verification confirmed physical consistency in 22 of 25 limiting cases, and numerical responses aligned with analytical reference responses in 30 of 32 grid configurations (Figure 2). The remaining warnings identify near-well finite leakage and coarse-grid settings that require caution before field transfer. Those warnings delimit the usable benchmark domain.

The screened benchmark separates response fit from parameter recovery. Across class and interpretation groups, the median model-factor COV values are 2.06 for transmissivity, 1.27 for storage, and 1.21 for response time (Figure 3). Leaky, heterogeneous leakage, and combined boundary classes have wider upper tails. Variance attribution shows that response residual standard deviation is driven mainly by water-level noise and observation support, whereas transmissivity and storage model factors remain strongly interpretation dependent. A small drawdown residual does not guarantee that the inferred T or S is close to its support-matched reference value.

3.2. Benchmark-conditioned transfer to field diagnostics

Benchmark-transfer regressions predicted the 90th percentile absolute log model factor with cross-validated R^2 values of 0.48 for transmissivity, 0.51 for storage, and 0.55 for response time (Figure 4). These moderate values establish a reproducible bridge from response diagnostics to conditional parameter uncertainty. Applied to the field cases, the p90-equivalent benchmark-transferred model-factor COV values are 0.61–0.90 for Massachusetts and 0.31–0.51 for Lovelock Valley. The values act as sensitivity brackets conditioned on the represented benchmark classes and field diagnostics.

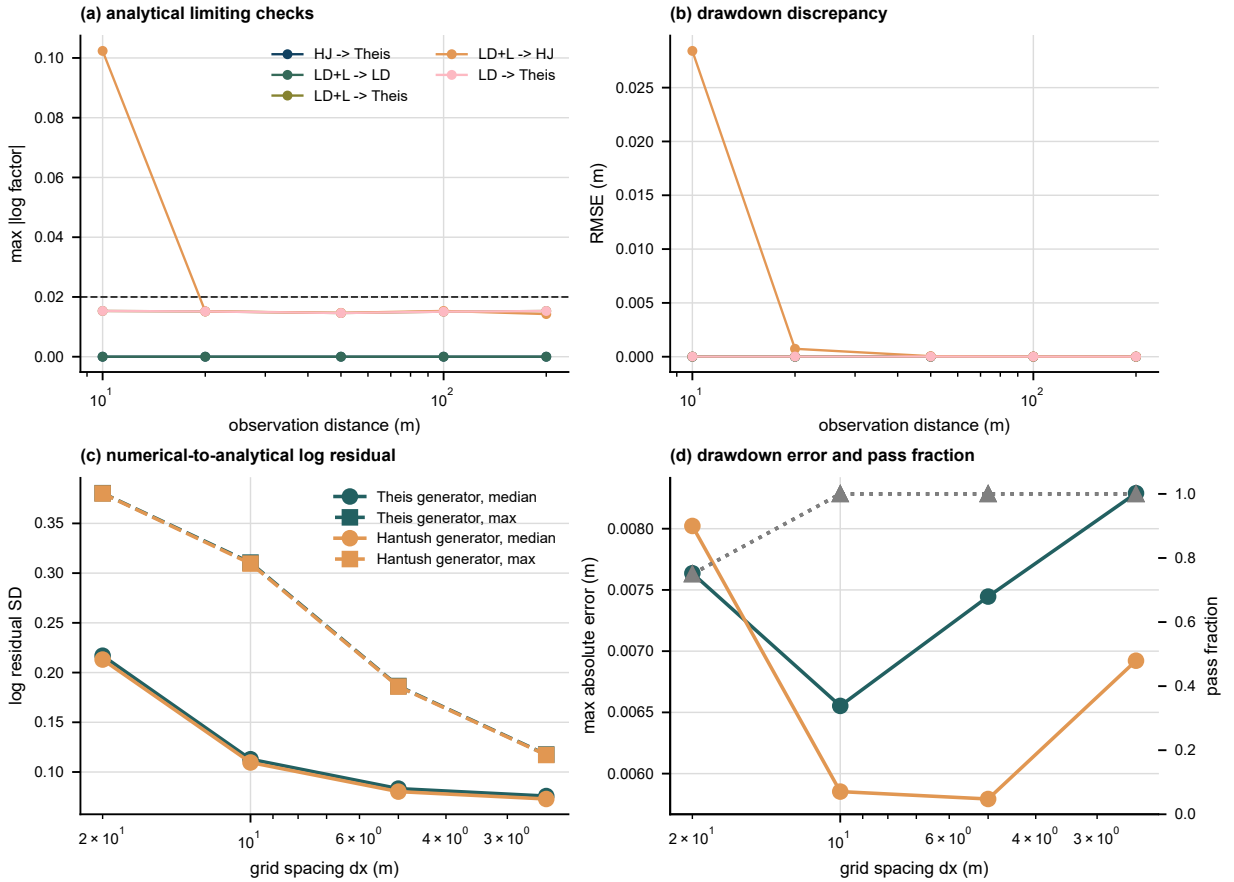


Figure 2: Analytical verification and numerical consistency checks through (a) analytical limit behavior versus observation distance and the dashed log factor tolerance for limit checks; (b) additional analytical limit behavior versus observation distance; (c) numerical consistency with homogeneous Theis analytical reference responses across grid intervals; and (d) numerical consistency with vertical leakage Hantush analytical reference responses across grid intervals. MODFLOW 6 numerical responses and caution labels define finite leakage, near-well, and coarse-grid limits for benchmark scope.

3.3. Field diagnostic evidence

The two field cases place the benchmark in real aquifer-test settings with different geological supports (Figure 5). Massachusetts is a small fractured-bedrock network, whereas Lovelock Valley is a basin-fill system with pumping and recovery observations. The four analytical interpretations were fitted consistently across both cases (Figures 6 and 8). The visual fits and residual summaries show that the lagging interpretations reduce structured response mismatch in both settings, while leakage support differs by case and well.

In Massachusetts, pooled log model factor dispersion decreases from 0.166 for Theis confined to 0.062 for lagging Darcy without leakage and 0.035 for lagging Darcy with leakage (Figure 7). Hantush–Jacob and lagging leaky forms receive leakage support in selected wells, indicating that leakage or boundary-equivalent behavior should remain an explicit competing interpretation. In Lovelock Valley, pooled dispersion decreases from 0.246 for Theis confined to

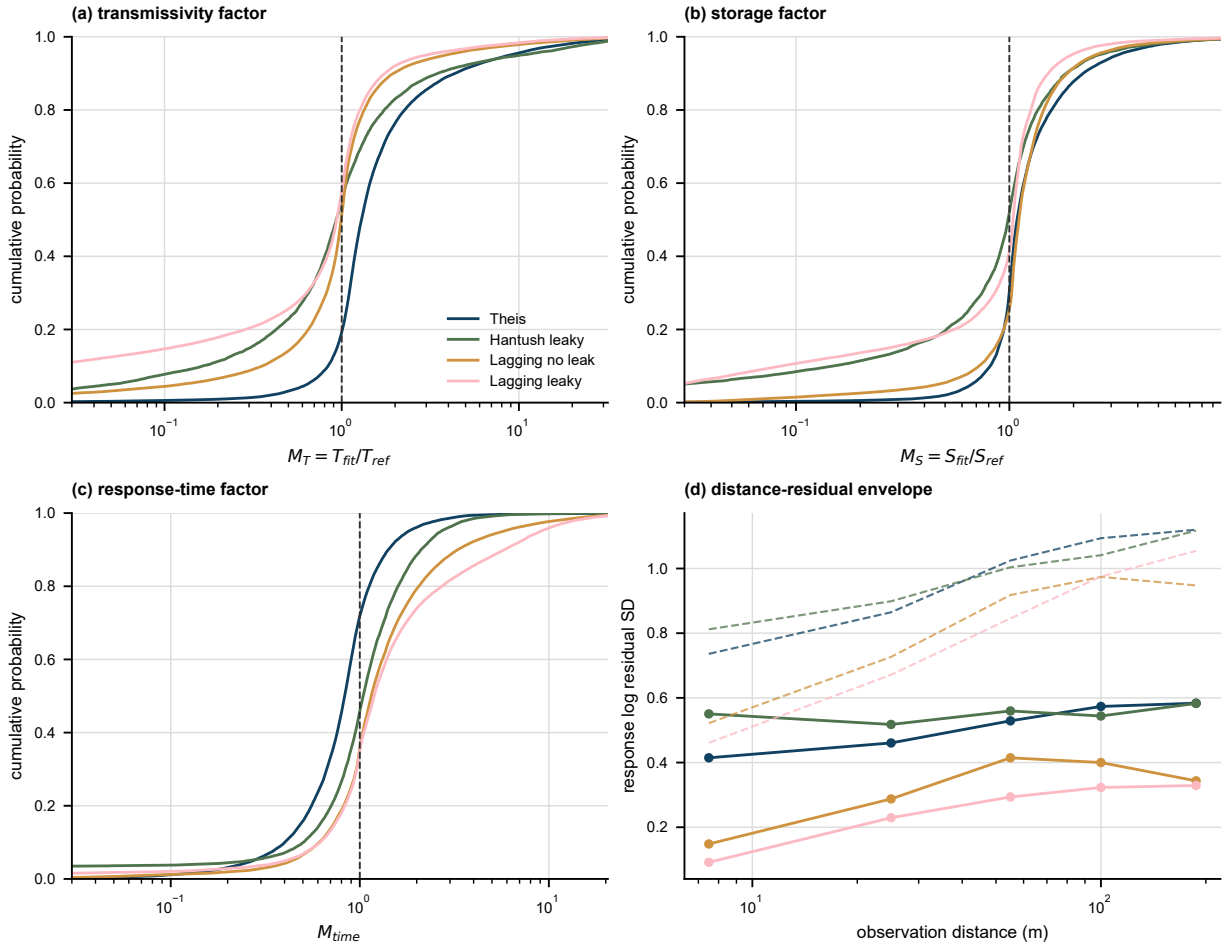


Figure 3: Screened numerical benchmark model factors through (a) empirical distribution functions for transmissivity model factors after boundary-active fit removal and retention of interpretations within $\Delta BIC \leq 10$; (b) empirical distribution functions for storage model factors under the same screened records; (c) empirical distribution functions for response-time model factors under the same screened records; and (d) median response log residual standard deviation by observation distance, with dashed lines for the 90th percentile envelope. The vertical dashed line in (a)–(c) denotes a factor of one, and all values represent scenario-conditioned model factors for the benchmark classes.

approximately 0.174 for the two lagging interpretations, while BIC support concentrates in the no-leakage lagging form (Figure 9). The site contrast matches the intended use of the four pathways: conventional leakage, temporal memory, and combined behavior are tested rather than assumed.

3.4. Scope of inference and management consequence

The synthesis figure separates the inference scope of the two field cases (Figure 10). Both cases contain response-level diagnostics and pathway-dependent parameter spread. The Massachusetts case includes a within-network spatial-transfer diagnostic by excluding one observation well at a time; Lovelock adds an independent field consistency check under a different support scale. Neither field case by itself recovers true support-scale T , S , leakage, or lagging times.

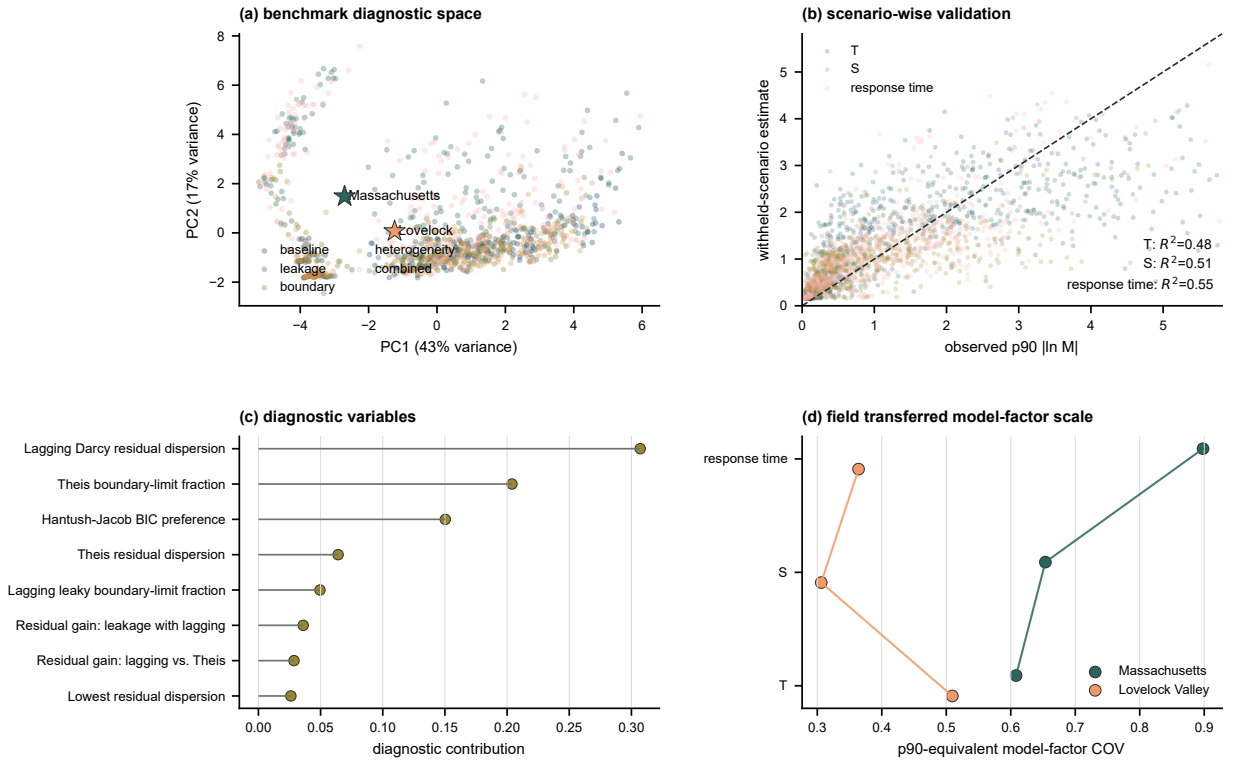


Figure 4: Benchmark transfer of aquifer-test transformation uncertainty to field diagnostics through (a) the numerical benchmark response-diagnostic space and the positions of the two field cases; (b) cross-validated predictions versus observed 90th percentile absolute log model factors for transmissivity, storage, and response time; (c) the main diagnostic variables for the regression; and (d) benchmark-transferred, 90th-percentile-equivalent model-factor COV values for the two field cases.

The numerical benchmark supplies conditional model factors, and the field applicability criteria define where those factors can be used as sensitivity brackets or conservative priors (Figure 11).

The decision diagnostic shows why this upstream uncertainty matters for later groundwater calculations. Under variance-window BMA, the median hydraulic-capacity factor is 1.60 in Massachusetts and 1.18 in Lovelock Valley; the corresponding 95th percentile factors are 2.73 and 4.52 (Figure 12). Median response-time factors are 1.85 and 4.05, with 95th percentile values of 3.32 and 8.79. These diagnostic inheritance factors differ from project-specific design factors. They indicate that groundwater calculations using a single interpreted T and S can become overconfident if the aquifer-test report does not state the interpretation model, residual structure, parameter spread, benchmark support, and field-to-benchmark applicability.

3.5. Implications and limitations

The computational benchmark changes the role of field pumping tests in uncertainty analysis. Reported transmissivity and storage are treated as products of an interpretation step, so the workflow records the transformation

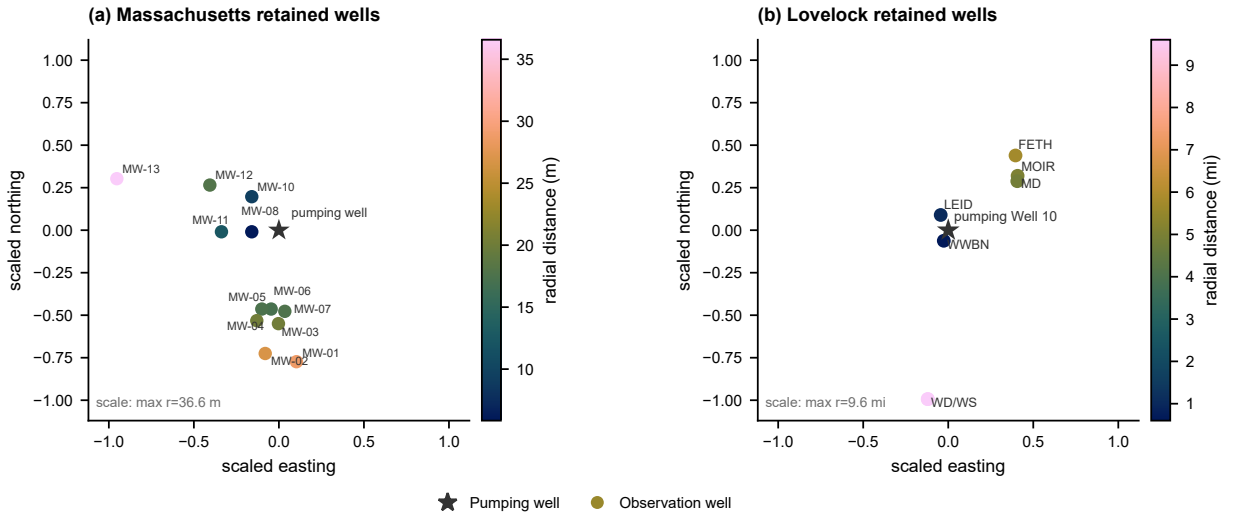


Figure 5: Paired field-data spatial support for the two real cases through (a) Massachusetts fractured bedrock retained observation wells; and (b) Lovelock Valley MWP2 retained observation wells relative to extraction Well 10. Both panels use coordinates centered on the extraction well and normalized by the maximum retained radial support within the case, so panel shape denotes observation geometry rather than absolute field scale. Circles denote retained observation wells, stars denote extraction wells, and colors retain the actual radial distance within each case.

that produced them. This separates response residuals, parameter model factors, and inherited decision factors. The distinction complements stochastic hydrogeology and groundwater BMA because later models can only propagate uncertainty that is represented in their inputs.

The evidence also defines the applicable domain. The field COV values apply to the two data sets and to the four implemented interpretations. Benchmark model factors are conditional on the scenario class, support definition, observation geometry, pumping schedule, grid design, and screening criteria. Direct evaluation of true field parameter recovery would require independent support-scale measurements. Broader model families remain future extensions, including wellbore storage, skin, delayed yield, generalized radial flow, dual porosity, and unconfined numerical benchmarks. Within that domain, the present result is a reproducible reporting standard: aquifer-test parameters should be accompanied by response model factors, complexity support, pathway-dependent parameter spread, benchmark applicability, and decision inheritance before they are used as deterministic groundwater model inputs.

4. Conclusions

This study presents a CAGEO-oriented computational benchmark for aquifer-test transformation uncertainty. The workflow links measured drawdown, four analytical interpretation pathways, response-level log model factors, support-matched numerical reference values, benchmark model-factor distributions, field applicability criteria, and model-averaged decision diagnostics. The 10,000 scenario benchmark produced 160,012 fitted interpretation records; analyt-

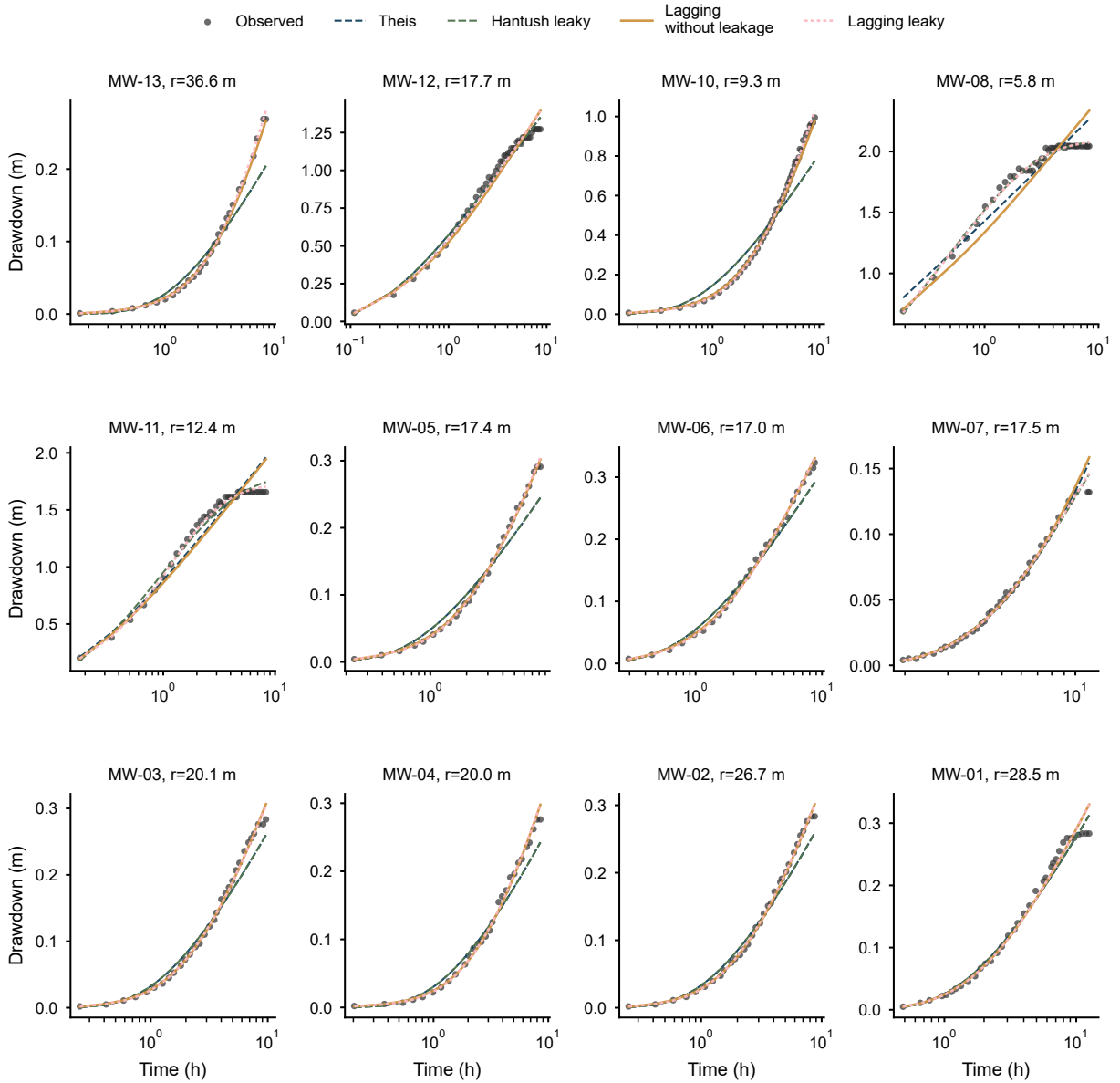


Figure 6: Massachusetts observed and simulated drawdown curves for the 12 observation wells across the four main interpretation models, with measured drawdown as points and Theis confined, Hantush–Jacob leaky, lagging Darcy without leakage, and lagging Darcy with leakage predictions as lines.

ical and numerical checks confirmed 22 of 25 analytical limiting cases and 30 of 32 numerical consistency configurations. The results separate response fit, parameter recovery, and decision inheritance.

The two field cases demonstrate the use of the benchmark. Lagging Darcy interpretations reduce response residual structure in both Massachusetts and Lovelock Valley, but leakage support, parameter convergence, and field transfer differ by case. The benchmark-transfer analysis gives conditional model-factor COV values rather than universal design

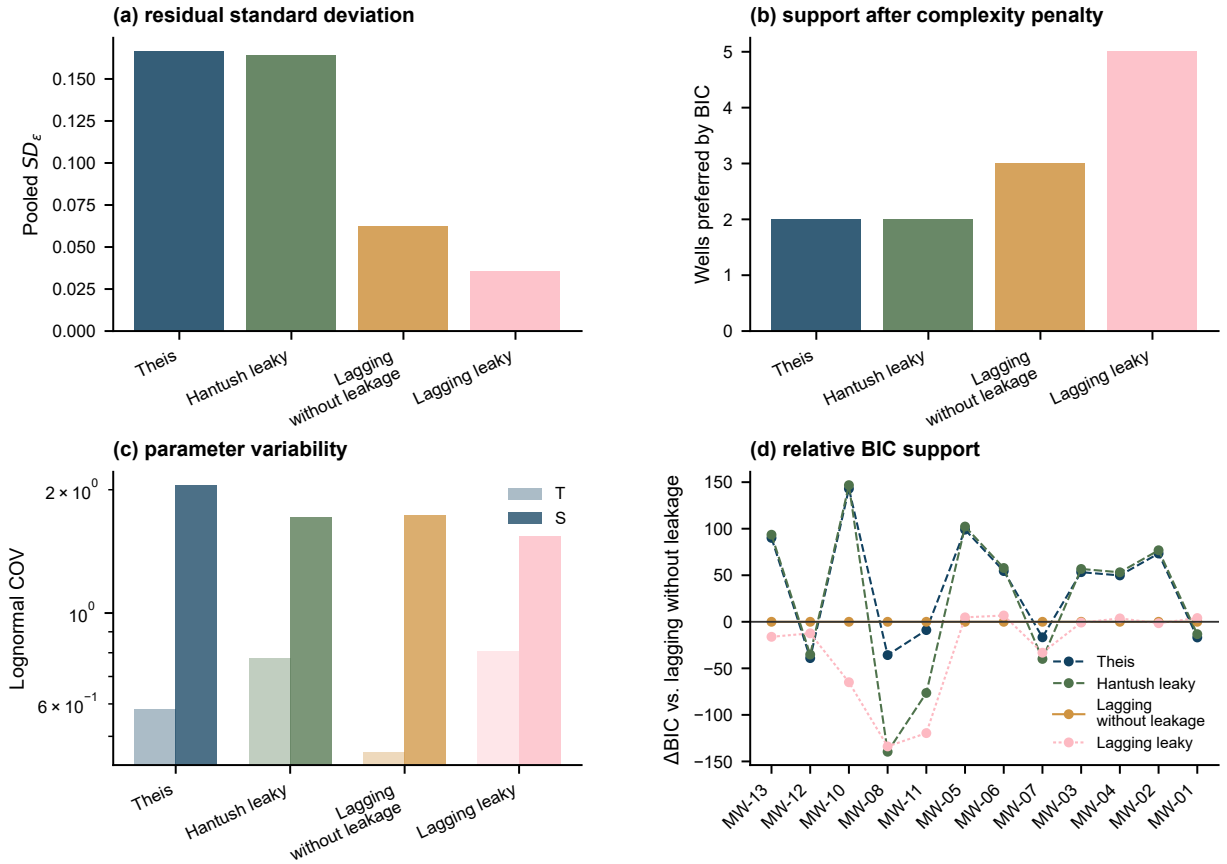


Figure 7: Massachusetts interpretation-model summary through (a) pooled log model factor dispersion; (b) BIC preference counts across the four main interpretation models; (c) lognormal parameter COV for fitted transmissivity and storage; and (d) relative BIC support against lagging Darcy without leakage.

values. The management diagnostic shows that collapsing interpretation uncertainty into one fitted parameter set can understate hydraulic capacity and response-time spread. The main contribution is a reproducible code-backed workflow for reporting how aquifer-test drawdown becomes groundwater model input.

Acknowledgments

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Computer Code Availability

Name of code/library: Aquifer-test transformation uncertainty benchmark workflow.

Developer and contact: Ying-Fan Lin, Department of Civil Engineering, Chung Yuan Christian University,

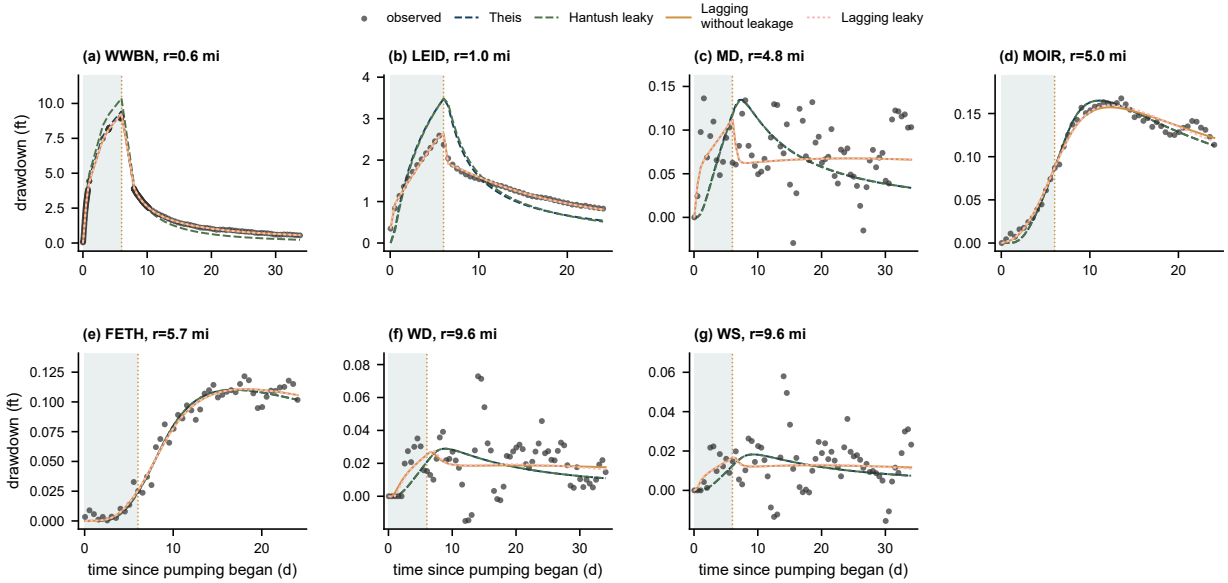


Figure 8: Lovelock Valley MWP2 observed drawdown and recovery responses with fitted curves under the four main interpretation models across seven selected observation wells by distance from Well 10, with measured responses as points, four model predictions as lines, the shaded interval as the extraction period, and the dotted line as shutoff.

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The source code is openly available under the MIT License at <https://github.com/aar246860/aquifer-test-tu-benchmark>. The workflow uses Python 3 and LaTeX. Python dependencies include NumPy, pandas, SciPy, scikit-learn, matplotlib, cmcrameri, FloPy, and Pillow. MODFLOW 6 is required only for rerunning the full numerical groundwater-flow simulations; the repository quick test does not rerun the 10,000 scenario benchmark. A lightweight reproducibility check can be run with `python examples/quick_test.py`. The larger generated benchmark and field-analysis tables are archived in HydroShare at <https://www.hydroshare.org/resource/3e8de7ff140c4ffab03163e215bb6634/>.

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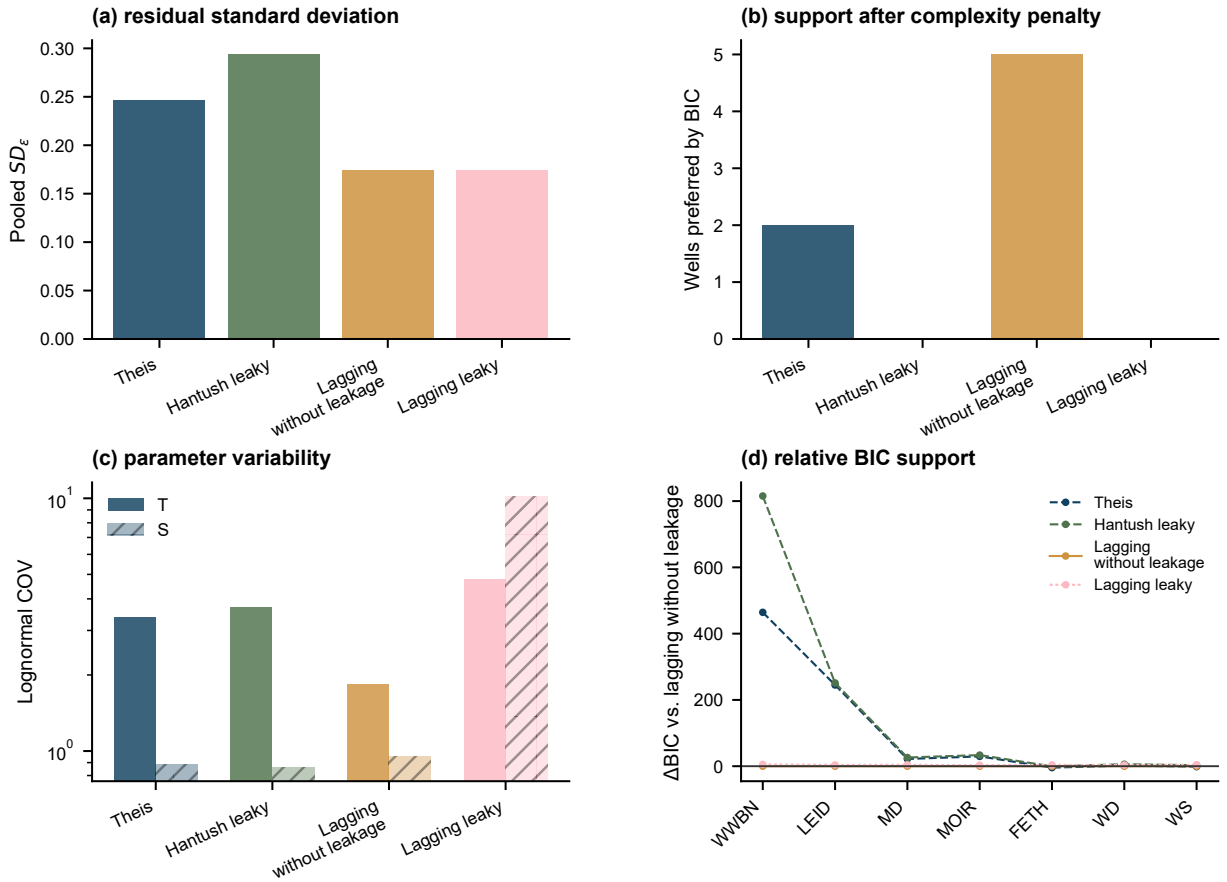


Figure 9: Lovelock Valley interpretation-model summary through (a) pooled log model factor dispersion; (b) BIC preference counts across the four main interpretation models; (c) lognormal parameter COV for fitted transmissivity and storage; and (d) relative BIC support against lagging Darcy without leakage.

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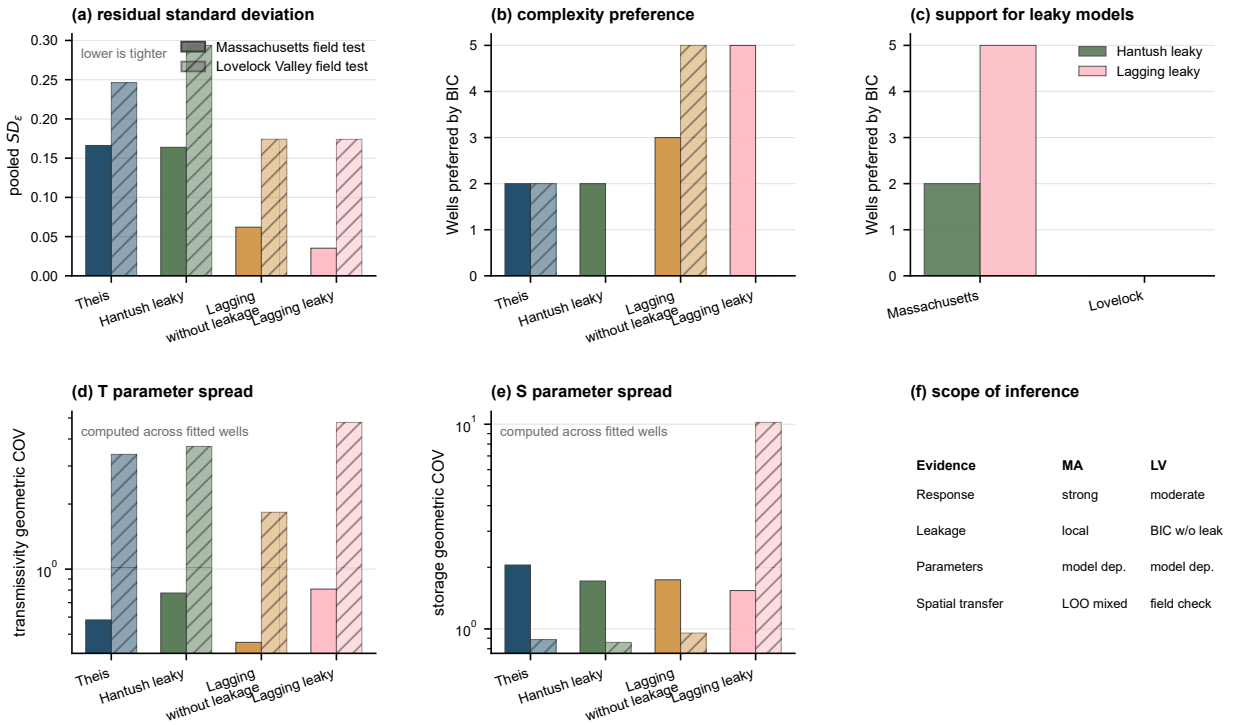


Figure 10: Validation synthesis from the Massachusetts fractured bedrock case and the Lovelock Valley basin-fill case through (a) pooled log model factor dispersion; (b) BIC preference counts by well; (c) support for leakage-inclusive models; (d) transmissivity COV; (e) storage COV; and (f) inference scope across response support, leakage support, parameter interpretation, and spatial transfer availability.

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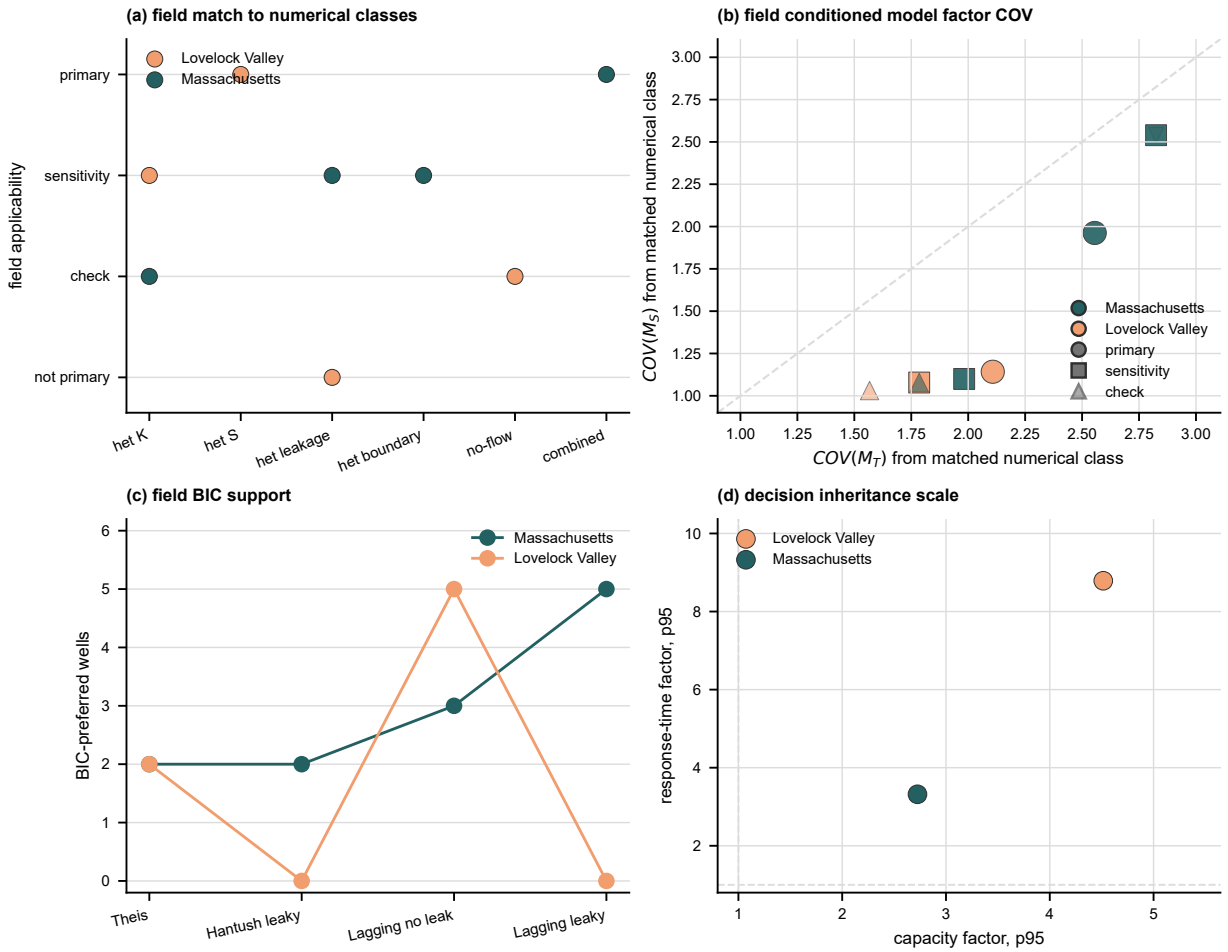


Figure 11: Field application of scenario-conditioned transformation model factors using the field evidence chain through (a) assignment of the two field cases to numerical benchmark classes and applicability levels; (b) matched transmissivity and storage model factor COV values as field sensitivity brackets or conservative priors; (c) field BIC support for the four analytical interpretations; and (d) inherited 95th percentile capacity and response-time factors from the model-averaged decision diagnostic. The figure identifies where benchmark factors enter the field interpretation.

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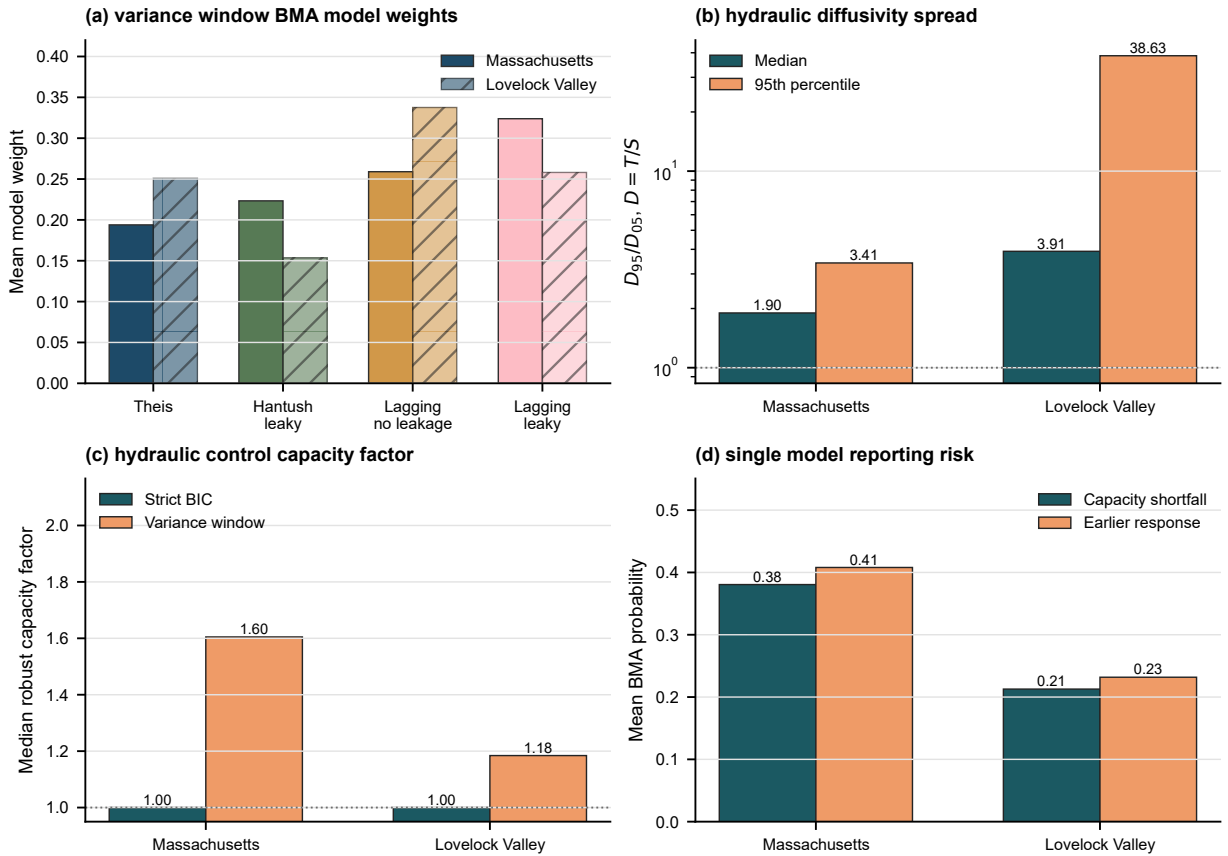


Figure 12: Model-averaged propagation of aquifer-test interpretation uncertainty to management-relevant hydraulic quantities through (a) mean variance-window BMA weights for the four analytical interpretations; (b) median and 95th percentile hydraulic diffusivity spread, expressed as D_{95}/D_{05} with $D = T/S$; (c) median hydraulic-control capacity factor under strict BIC weights and variance-window weights; and (d) mean variance-window BMA probability of selected-interpretation underestimation of hydraulic control capacity or missed earlier hydraulic response.

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